Annexure – 5

CONTRACT SAFETY DOCUMENT

SAFTEY GUIDE LINES

1. Introduction:

1. This guide is applicable for all site contractors, sub-contractors and vendors working at projects owned and/or leased by GENERAL CONTRACTOR for all activities on the construction site, including the applicable safety regulations (if any) – as enforced time to time by statutory bodies.

   1(a) for safety any cost related to safety at site as directed by the construction manager and not limited to the items listed shall be borne by the NOMINATED SUBCONTRACTOR.

   1(b) The NOMINATED SUBCONTRACTOR shall at his own expense arrange or complying with all the occupational Safety, Health, Environment and welfare legislations of prevailing for respective states and central Government of India wherever applicable.
2. Instructions issued herein are indicative and not exhaustive .Therefore the NOMINATED SUBCONTRACTOR shall be responsible to ensure that adequate safety measures has been adopted in the course of execution of the contract in accordance with all applicable rules and regulation of legislations applicable to the construction activity.
3. GENERAL CONTRACTOR Safety management system (Construction Safety Manual), formats & documents available at GENERAL CONTRACTOR site office. NOMINATED SUBCONTRACTOR needs to refer all safety guideline & follow during day to day activities at site.
4. The NOMINATED SUBCONTRACTOR shall implement safety measures as suggested and discussed with GENERAL CONTRACTOR & consultant from time to time to safe guard the employees from injury apart from the measures given in their safety management plan.
5. NOMINATED SUBCONTRACTOR shall ensure all contract safety rules and regulations followed by them and any sub-contractors, vendors, visitors etc. appointed by them.
6. More details of safety rules and regulations shall be stipulated by site management by means of supplements to the site regulations.
7. GENERAL CONTRACTOR's Project Manager shall have the right to stop the work at site, if in his opinion proceeding with the work will lead to an unsafe and dangerous condition.
8. If NOMINATED SUBCONTRACTOR violates any safety, rules and regulations on site will be penalized as per GENERAL CONTRACTOR's safety non-compliance regulation.
9. The NOMINATED SUBCONTRACTOR shall be solely responsible for any non-compliance of any of the safety measures and delays, implications, injuries, fatalities, damage to properties and compensation arising out of such situation or incidents.
10. The NOMINATED SUBCONTRACTOR shall take all safety precautions during the execution of awarded work and shall maintain and leave the site safe at all times.
11. The NOMINATED SUBCONTRACTOR shall ensure that all workmen are aware about the nature of risk involved in their work and have adequate knowledge, training for carrying out their work safely.
12. It is all parties/agencies responsibility endeavor to secure a high standard of safety on our sites. Therefore, NOMINATED SUBCONTRACTOR and sub-contractors must know their duties under all applicable relevant statutory laws, both for establishments and their own

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employees and to conduct their business and methods of work they shall be carry out to confirm to the best practices.

2. Induction training and screening of workmen:

- The NOMINATED SUBCONTRACTOR shall ensure that all employees give safety orientation training before engaging them on job and workmen are screened.
- All employees engaged by the NOMINATED SUBCONTRACTOR should be medically fit for the job they are entrusted with.
- Physical fitness of the person to certain jobs like working at height, confined space, drivers /operators or engaged in other dangerous operations/activities to be ensured before engaging the person on work.

3. Information, Briefing, Training:

- The personnel must be briefed by their employer when the work begins and at regular intervals, to the effect and risks deriving from their activities as well as on steps taken for their prevention.
- All workers receive specific task related training.
- The site management may stipulate the intervals for such Tool box talks.
- These briefing shall be documented.

4. Site Access, transport & visitors on the site

- The NOMINATED SUBCONTRACTOR shall organize operations by taking approval to use the existing roads, if any from employer.
- The NOMINATED SUBCONTRACTOR shall exercise full care to ensure that no damage is caused by him or workmen, during the operation, to the existing water supply, sewerage, power or telecommunication lines or any other services or works. The NOMINATED SUBCONTRACTOR shall provide and erect before construction, substantial barricades, guardrails, and warning signals. He shall furnish, place and maintain adequate warning lights, signals etc. as required by GENERAL CONTRACTOR.

Entering or leaving the site:

- The site shall be entered or left only at the construction site main gate.
- All persons and /or vehicles entering or leaving the site, including objects or materials they carry with them and any locker on site are subjected to search or other control measures by the security staff authorized by GENERAL CONTRACTOR's site management.
- Waiting on the site is permitted to suppliers only for the orderly execution of deliveries.

Transport to the site:

- All supplies must be marked clearly and visibly and be accompanied by appropriate documents.
- The recipient has to ensure that transport; handling, provision or storage of the shipment is carried out orderly and correctly without endangering third persons, the plant, the site equipment or the environment.
- The goods which are considered as being dangerous must be reported to the site management by the recipient prior to bringing on site.

Material gate pass:

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- Materials, equipment, tools, etc. may be hauled away from the site only upon approval by the GENERAL CONTRACTOR's site management.

Control measures:

- The transport to the site and the hauling away of materials, tools, etc. are subject to control measures ordered by the site management according to above section.

Private Vehicles:

- Driving on the site with private motor-vehicles, motorcycles and bicycles is not permitted unless authorization by the GENERAL CONTRACTOR's site management.
- Private vehicles have to be parked in an orderly manner in specified areas.

Road traffic rule:

- The rules of the road traffic acts apply to the vehicle traffic within the site area. The vehicles must be in a safe condition, insured against liability. Maximum speed limit is 10 KM/Hr.
- For the use of cranes, tipper Lorries, and other heavy vehicles operating in proximity to overhead power lines is not allowed until specific permission given.
- All vehicles must be clearly identified with the name of the NOMINATED SUBCONTRACTOR it is used by.
- Gross pollution of or damage to the roads must immediately be removed by its originator at his expense. If the originator is not carrying out the cleaning or the repairs in due time the site management may restore the original conditions at the expenses of the originator.

Access to the site and working areas:

- Access shall be kept free at all times. In case of potential obstruction, due to interference between erection activities and other activities these shall be brought to the site management's attention for a proper solution, prior to its occurrence.
- Adequate and safe means of access and exit shall be provided for all work places, at all elevations.

Additional access ways:

- Access which may be required in addition to existing site roads can be installed only upon agreement with site management. This authorization must be obtained from GENERAL CONTRACTOR's site management.

Escape, Traffic and Transport Routes:

- All routes, especially escape, traffic and transport routes must be kept clear.
- Any obstruction or danger caused by material, scaffoldings, cables lying on the ground, and pipes, is generally prohibited.
- Major obstructions occurring in unpreventable individual cases require the prior agreement of the site management, in order that substitute arrangements may be laid down presented upon request.

5. Photographing, Video recording and shooting films:

- Photographing on the site is generally not permitted .Authorization must be previously obtained from the GENERAL CONTRACTOR's site management.

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6. Visitors:

- Visitors are not permitted to visit operational areas. Special permission shall be obtained from GENERAL CONTRACTOR's site management. When visiting operational areas must be accompanied at all times by a person who has undergone safety induction training.
- Visitors who allowed for visiting project site must use Safety helmet, Safety shoes, reflective jacket and required PPE.

7. Site Entry –Workers:

- All workmen shall have valid Identity badges issued by the NOMINATED SUBCONTRACTOR project manager or as directed by Project Manager GENERAL CONTRACTOR's.
- Entry only with Safety helmet, safety shoes and reflective jacket.
- For Visitors PPE will be issued before entry and collected back prior to their exit.

8. Site management's Authority to Issue Instructions:

- The GENERAL CONTRACTOR's site management is authorized to issue instructions to all contractors and their employees on site to meet his contractual requirements, observing the laws, ordinances etc.in force.

9. Co-operation with the site Management:

- NOMINATED SUBCONTRACTOR, each contractor shall fully co-operate to implement safety at site and maintain Zero injury /accident policy at site.

10. Safety Meetings:

- The contractors shall attend all safety meetings weekly as instructed by the GENERAL CONTRACTOR's site management.
- Safety compliances, safety performances and other safety related issues shall be reviewed in this meeting for further improvement. NOMINATED SUBCONTRACTOR's project manager, site in charge and site engineers along with safety manager shall be present for safety meeting.

11. Personnel/refusal of personnel:

- The NOMINATED SUBCONTRACTOR's personnel employed on the site must be personally and professionally qualified for the duties assigned to him. The employees of the various companies are, in addition, obliged to good and smooth co-operation with the personnel of other companies.
- Personnel that disturbs peaceful labour relations, transgress accident prevention regulations or do not obey instructions issued by the site management shall be subject to disciplinary action. Costs incurred thereby shall be paid by the NOMINATED SUBCONTRACTOR involved.
- Site management has the right to request the removal from site of any person(s) that it considers to be not suitable.

12. Use of erection equipment belonging to other contractor/agencies.

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- Good will of co-operation and assistance between the NOMINATED SUBCONTRACTOR is obligatory e.g. giving possibility in using erection equipment, available on site and belonging to other contractors.
- In each case the user will take the full responsibility for the proper handling of the borrowed equipment. This also applies to the fixed mounted lifting devices.
- The GENERAL CONTRACTOR of the equipment is responsible for ensuring its satisfactory state of maintenance, and, where applicable, statutory inspections (Third party Inspection). The GENERAL CONTRACTOR must also ensure that the user is given adequate instruction on the safe use of the equipment.
- Drivers of driver-operated cranes nominated by the crane GENERAL CONTRACTOR must have valid license and competency certificate.

13. Site facilities:

General:

- Before erection of any temporary site facilities like office, access routes for temporary structures, labour camp, roads, stores cranes, services, etc. NOMINATED SUBCONTRACTOR smust obtain approval from the site management of the size, type, condition, location and services. The site management approval shall be obtained before any facility is dismantled or removed from the site.
- The NOMINATED SUBCONTRACTOR `s facilities shall be reasonably identified with the NOMINATED SUBCONTRACTOR's name.
- The NOMINATED SUBCONTRACTOR shall bring the information necessary to reach the approval in good time with the aid of a site facility plan, in order to ensure all the preconditions required for the work to begin on schedule.
- The site management will allocate to each contractor an appropriate area for the erection of temporary offices, labour camp, and storage or prefabrication area, facilities. It is the NOMINATED SUBCONTRACTOR responsibility to prepare the area, including walkways for his personnel, to suit his requirements at his own expense. It is the responsibility of the NOMINATED SUBCONTRACTOR to connect to these terminals in accordance with statutory requirements at his expense.
- The legal stipulations in force must be respected; the same applies to a NOMINATED SUBCONTRACTOR.
- The NOMINATED SUBCONTRACTOR is responsible for the payment of all applicable local taxes, rates, etc.
- All temporary buildings, labour camp and facilities must be dismounted and hauled away upon termination of erection or commissioning ,foundations and roads shall be removed without delay and the area concerned be left in an orderly state.

14. NOMINATED SUBCONTRACTOR's Facilities:

- The NOMINATED SUBCONTRACTOR's must mandatory maintain first-Aid room at site ,additional medical facilities, Health care, pest control, waste disposal, welfare facilities like Potable filtered drinking water , sufficient Toilets and urinals shall be made available for the work force at various locations at site along with good hygiene & house-keeping with an approximate ratio 1: 20 and , rest room, rain shelters, mess rooms, sanitary conveniences and washing facilities, according to the number of their employed personnel and general standards. Exceptions have to be settled in separate written agreement.

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- The site management reserves the right to reject any of the NOMINATED SUBCONTRACTOR's facilities if it not fulfills the relevant statutory regulations or site standards.
- Where a NOMINATED SUBCONTRACTOR fails to install and maintain a satisfactory standard of welfare facilities for his work force, GENERAL CONTRACTOR reserves the right to install and maintain such facilities at the NOMINATED SUBCONTRACTOR's cost.
- Unauthorized stay at site not permitted.

15. Temporary camps:

All sites used for temporary work camps permitted following requirements.

- All sites used for temporary work camps shall be properly drained.
- There shall not be subject to periodic flooding nor located within or in vicinity to area water logging ,sink holes, swamps, improper ventilation and surface collection of water unless adequate mosquito control measures have been implemented.
- Overcrowding shall be prevented. Develop identical size of rooms at labor camp with sufficient passages, in and exit routes, etc.
- Open areas, grounds surrounding the shelter shall be maintained free from debris, waste paper, rubbish, combustible material, garbage and other refuge material, grass, etc.
- Camp area shall be protected from other elements. Take adequate snake preventive/repellent precautions.
- Suitable flooring shall be provided of smooth and tight construction. Floors shall be kept in good repair.
- Temporary camps shall be ventilated, lighted and maintained in a clean and safe condition.
- Camps must comply with the local authorities fire protection requirements.
- All welfare & sanitary facilities shall be provided.
- To control & monitor labor work camp security & camp boss shall be provided.
- To control risk of fire, provide fire water tank with separate fire pump provision.
- Adequate area lighting/passage lighting shall be provided.
- The designing of the labour camp main electric panel & electric distribution should be done by trained & qualified Electrician only.
- Electric panel with the adequate capacity should be installed for labour camp power supply.
- In hutment areas the electric wiring /cabling to be done through conduits.
- All electric connections through electric distribution board and distribution board should have ELCB of 30 mA cut off ratings and earthing continuity from earth pit.
- Heavy current drawing equipment's such as oven, hitters etc. should not be allowed to use.
- Electrician should regularly check/test earth pit, ELCB, & cable joints insulations.
- Adequate Common area lighting arrangement to be provided to enable safe access /egress of workers at labor camp, toilets, water tank, access roads, etc.
- Adequate number of firefighting equipment should be kept in strategic position and laborer & their family member should be train in firefighting.
- If workers are staying on site in Labour camp Wood burning is not allowed.
- Provide separate potable drinking water, washing water provision with adequate foundation and water drainage system.
- Periodic (monthly) checking of drinking water sample by third party approved by MPCB.
- To control & monitor labour camp, deploy camp supervisor & security personal.

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- Deploy sweeper for daily cleaning of toilets, common areas, access/egress road, labour camp surrounding area cleaning, collection, disposal of food waste, garbage waste and maintaining hygienic conditions in labor camp.
- Provide covered dust bin.
- Garbage /waste food collection yard should be set up away from living area.
- Daily disposal of food waste and garbage waste to be carried out safely.
- Provide proper fencing surrounding labor hutment.
- Adequate water drainage facility shall be provided to prevent water logging, accumulation of water and to eliminate monsoon water rains.
- Pest control: Periodically carry out Fogging, spraying of mosquito crude oil, pest control, snake repellent spray, rats and rodent control measures.
- Staying of women(s) and children(s) in labour camp is prohibited.
- Cooking inside room is not allowed, separate cooking area should be provided.
- NOMINATED SUBCONTRACTOR shall appoint visiting doctor (weekly) for checking & providing medical treatment for illness of people staying in labor camp.

16. Fire Protection:

- NOMINATED SUBCONTRACTOR, each contractor must ensure that a sufficient number of appropriate operable firefighting equipment, sufficient water storage/supply are within easy reach in his office rooms, stores areas, workshops, building floors, electric panels, list of major workplace fire hazards, potential ignition sources etc.
- Adequate water supply at required pressure. NOMINATED SUBCONTRACTOR's scope of works excludes temporary firefighting arrangements. It is assume that permanent wet risers can be installed and used on the project during structure execution.
- The NOMINATED SUBCONTRACTOR to have a mock drill and training given to the labors/supervisors who will be deputed as fire watchers and the report of the same to be submitted to the P.M NOMINATED SUBCONTRACTOR on a monthly basis.
- Personnel shall be trained to operate the fire extinguishing equipment.
- Firefighting equipment shall be kept in good order.
- Provide adequate number of fire point at various locations with provision of fire extinguisher & dry sand bucket.
- The concerned NOMINATED SUBCONTRACTOR will be responsible for fire protection according to legal stipulations and contractual agreements.

The above rules will not relive any contractor / NOMINATED SUBCONTRACTOR or his personnel from his special responsibility and obligation when handling fire sources of any kind.

17. Buried Lines:

- Various kinds of buried cable and pipe exist on the site and in the general locality shall be protected .Damages to the lines may present danger and cause failure of important installations.
- Prior to the commencement of any excavation work , piling work the position of all underground services ,and method of excavation, is subject to the approval of the site management.
- The work in the vicinity of underground services, and method of excavation, is subject to the approval of the site Management.

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- Where the ground, weather or nature of the work is such that a collapse of earth may occur, suitable and sufficient shoring materials and supports must be provided as required by the statutory provisions.
- Due to danger to and from underground services the driving of any item into the ground is classified as an excavation. These works will be carried out at the expense of the initiator.
- To ensure that all staff and contractors are clearly aware of the risks of electric shock from excavation works and working between live overhead lines as detailed special permit system has been established and according to Indian electricity rule all required safety precautions to be taken.

18. Waste Disposal:

- Both organic and inorganic waste should be collected, separated & disposed as per the relevant state, local, municipal regulations without affecting the public & environment.
- No dangerous or noxious waste products, chemicals, hazardous waste or materials shall be disposed of, on or off the site without the site management's approval.
- Disposal of the waste doesn't harm to site people, general public and environment, dispose it safely according to the pollution control board regulations and meet local, state and federal waste management legislation and environmental standards, including waste minimization.
- Hazardous material waste shall be collected, stored, and disposed of in accordance with national, state and local applicable regulations.
- Waste material and rubbish shall be collected daily, placed in closed containers & disposed.
- In the event of an oil spill or similar environmental issue, the NOMINATED SUBCONTRACTOR responsible for the given condition or event is obligated to immediately rectify the pollution caused by appropriate cleaning actions and/or soil replacement.

19. Waste disposal by Burning:

- The burning of rubbish, oils and other waste material on the project is prohibited.

20. Workplaces:

- The NOMINATED SUBCONTRACTOR shall keep in an orderly condition their construction and working areas as well as stores and workshops etc. and shall, at least once a week, carry out a thorough cleaning action, records must be kept.
- If NOMINATED SUBCONTRACTOR fails to in his obligation to effect the required conditions, the site management may take such measures to remedy the conditions at the expense of the NOMINATED SUBCONTRACTOR. In the event of these measures being taken, the loss of tools, materials or other losses can't be claimed.

21. Aircraft Warning lights:

- The responsible NOMINATED SUBCONTRACTOR shall take all necessary aircraft warning measures on temporary installed lifting equipment, gear and on tall structures during their erection.

22. Construction site cleanliness on site in general:

General House-keeping on construction site:

- House-keeping and hygiene go in hand with safe working practices.

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- Throughout the period of construction the NOMINATED SUBCONTRACTOR shall maintain the area of his operation in a clean, tidy, and safe condition and shall arrange his material in an orderly manner.
- The NOMINATED SUBCONTRACTOR must ensure that he is not causing any pollution. Any gross defilement which the NOMINATED SUBCONTRACTOR or his sub-contractor causes to the roads as well as to the general accesses, staircases and completed rooms, areas or installed equipment of the plant must be removed immediately. Access and working areas must be kept free and in a proper and safe condition.
- Special attention shall be paid to potential fire hazards, trip points and equipment to ensure that they are not left in a hazardous condition.
- Access and working areas must be kept free and in a proper and safe condition.
- **GENERAL CONTRACTOR's standard Site Housekeeping process**: NOMINATED SUBCONTRACTOR shall follow following housekeeping process and implement on daily basis and maintain good housekeeping within site premises and at every workplace.

How do I plan a good housekeeping process?

We can adopt 5 Steps Housekeeping solution:

Housekeeping steps	Process details	Illustrations
Step 1	**Take out unnecessary items in workplace and keep them away.**	Work area: Removal of unnecessary construction debris, Material from  Storage of unnecessary material at identified "Debris collection area." 
Step 2	**Arrange necessary items in proper order**	Segregation of necessary material at workplace.  Orderly storage of necessary material at workplace. 
Housekeeping steps	**Process details**	**Illustrations**

DRAFT

Step 3	**Complete cleaning of your workplace**	Cleaning of work area  After cleaning keep all waste at Common disposal point. 
Step 4	**Maintaining housekeeping at workplace at all times**	Periodic cleaning of workplace to maintain housekeeping standard at  
Step 5	**Train people to follow and maintain good housekeeping disciplines**	 Create awareness among employee, contractor & their workforce to maintain good housekeeping and

23. Safeguarding against theft and loss:

- The NOMINATED SUBCONTRACTOR shall take sufficient precautions against theft and loss of his own objects as well as objects furnished by others.

24. Labor protection:

- NOMINATED SUBCONTRACTOR will be responsible to carry out all necessary steps for accident prevention and labor protection in accordance with the legal stipulations and contractual conditions.
- To prevent accident, appropriate facilities and arrangements must be provided and actions taken in accordance with health & safety at work legislation applicable to this type of work.
- Any representative of the GENERAL CONTRACTOR's site management will have authority to stop the work if, in his opinion, it is necessary in the interests of safety.
- If one working team performs his work at the same time and place as another team, protective measures have to be taken immediately to avoid risks to others. If the protection

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aim cannot be attained or other persons would be considerably impeded, a decision must be obtained from the site management.

25. Safety representative:

- Experienced safety staff with adequate number of supporting stewards shall be appointed on site during the execution of works by NOMINATED SUBCONTRACTOR.
- The appointed person shall also responsible for ensuring compliance with all appropriate statutory requirements and for promoting a safe working environment.

26. NOMINATED SUBCONTRACTOR's Obligations for health & Safety:

- NOMINATED SUBCONTRACTOR /subcontractors engaged for the contract works shall submit the following documents to the GENERAL CONTRACTOR's site management prior to commencing work on site.

1. Project Specific Safety Management Plan shall developed, implemented and updated as necessary, which is mandatory.
2. The relevant Safe work method statement s of all the key activities as well as risk assessment studies (HIRA`S) shall be attached to the safety plan (to be supplemented by additional necessary documentation as the works progress have to be prepared by the NOMINATED SUBCONTRACTOR and submitted to the site management.) provision of resources or safe working like equipmnets, PPE`s, etc.
3. Statutory Inspection Certificates for lifting machines, cranes, power tools, scaffoldings, Air compressors and receivers ,if any
4. Names of all qualified and approved operators of heavy machines, cranes, Explosive, Power Tools, and other skilled trades including Licensed Electrical worker. lifting supervisor and scaffolding supervisor;
5. The signed Acknowledgement of Safety rules and regulations by NOMINATED SUBCONTRACTOR /sub-contractors form GENERAL CONTRACTOR's PM shall brief the NOMINATED SUBCONTRACTOR's project requirement prior to job commencement at the site. GENERAL CONTRACTOR's PM shall sign as witness on the NOMINATED SUBCONTRACTOR /subcontractor's acknowledgment form.
6. Statutory requirements on NOMINATED SUBCONTRACTOR, each contractor remain mandatory or unaffected by any measures taken by site management to promote good HSE on site.
7. Names of safety officer and supporting safety staff with their competency certificates.

27. Site Inspections:

- GENERAL CONTRACTOR's site management carries out site inspections, audits, surveys, etc.
- A relevant invitation in good time from the site management makes participation obligatory.
- NOMINATED SUBCONTRACTOR should comply all site inspections /audit observations/safety violations and take immediate corrective actions, preventive actions and submit compliance status within time frame to GENERAL CONTRACTOR's Project manager.

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28. Medical facilities:

First-Aid Station &medical facility:

- All contractors should provide a first-aid room, at site in a readily accessible place with all First-Aid appliances/medicines as per the legal acts and rules and shall be maintained in all times.
- Full time trained Qualified first-aider shall be deputed. First-Aid attendants shall hold valid certificate in First-aid from authorized institute.
- An injured person shall be taken to a hospital without loss of time .The NOMINATED SUBCONTRACTOR shall make an agreement with a nearest hospital/dispensary for rendering further treatment to the employees.
- The first-aid station consists of standard First-Aid box complying all content as per mandatory requirement in addition to this basic fill, additional requirements in additional requirements should be maintained on recommendation of health professionals.
- Prior to commencement of work, arrangements shall be made for medical facilities and personnel to provide prompt attention to the injured.
- For the treatment of injuries, qualified first-Aider (shall be available on site round the clock.
- For snake bite treatment the required Anti-venum should be kept in refrigerator at the first-Aid center or make necessary arrangement (tie-up) with nearby hospital.
- For health care contagious deceases should be effectively controlled.
- The NOMINATED SUBCONTRACTOR to arrange for a site visit (twice in a week) of a certified doctor to have periodical check-ups done of the labors, in the labour camp as per the requirements.
- Maintain health center/first-aid center in hygienic condition.
- Ensure safe disposal of medical waste.

Emergency transport facility:

- NOMINATED SUBCONTRACTOR shall arrange a dedicated Ambulance facility with driver, for use only in emergencies, shall be stationed at site, whenever any kind of work is in progress at site to speedy care for injured workers.

29. Disaster Management Plan:

- The NOMINATED SUBCONTRACTOR must have disaster management plan available with him to deal in case of emergency situation. The plan should mention potential hazards of project site.
- NOMINATED SUBCONTRACTOR should submit Disaster plan to GENERAL CONTRACTOR's for his review & approval.
- Periodically conduct mock drill according to the plan to refresh the worker in emergency situation and also train new employees and any review of the training for any shortcomings should be accordingly modified and report should be submitted to GENERAL CONTRACTOR's.
- NOMINATED SUBCONTRACTOR should form emergency rescue team and responsibility should be assigned to them & to be prominently communicated & displayed at site.
- An effective means of emergency communication system should be available and established. Emergency warning device or siren shall be installed.
- Record of emergency drill should be documented.

DRAFT

- A dedicated space to be created at the site at suitable locations and the same to be marked as Assembly points, where in all the labors should assemble in case of emergencies. A display board showing Assembly areas to be erected at these said locations on site.
- All workmen on site should be made aware of the actions to be taken when any emergency arises on site.
- The list of addresses and telephone numbers of physicians, hospitals, or ambulances, emergency contact persons shall be display at relevant place which should be visible and accessible for any person.
- A map showing shortest route to the nearest medical facility/hospital shall be prepared and displayed.

30. Pest control:

- If pest control required on the site facilities and workplaces assigned to the NOMINATED SUBCONTRACTOR it must be carried out by himself at his own expense.
- Regular provision for pest control should be planned and carried out for control of decease causing pests.
- Mosquito control: effective control measures should be taken for control mosquito, insects, flies, rodent etc.

31. Registration and Notice of Accidents:

- All physical injuries/accidents /Near misses must be reported to the site management (GENERAL CONTRACTOR's P.M.) immediately on any occurrence.
- All records and legal notices of accidents must be carried out by the NOMINATED SUBCONTRACTOR himself.
- Calling the authorities in accordance with the pertinent regulations will be NOMINATED SUBCONTRACTOR , each contractor's duty.
- The health & safety officer /representative and the site management must be immediately informed in the event of serious or fatal accidents.

32. Interference with Existing protective and safeguarding facilities:

- The unauthorized altering of protective and safety facilities, in particular the removal of parts from protective and safety facilities is strictly prohibited and can lead to the dismissal from site of the person(s) involved.
- If protective and safety facilities have to be altered or removed as part of the work, the originator of the alteration or removal must make substitute arrangements and, if third persons are endangered, he must obtain the agreement of the site management. This applies in particular to removable coverings, gratings, barrications and the like.
- Cost incurred to install, re-fix existing protective and safe guarding facilities will be debited/recovered from NOMINATED SUBCONTRACTOR.

33. Safe guarding of Workplaces:

- The NOMINATED SUBCONTRACTOR is responsible for ensuring that the entire area of their construction and erection sites shall be safeguarded in such a way that there is not possibility of danger.

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- This must be observed particularly in the case of construction and erection work performed above pits which are open ,overhead work or floor openings covered by gratings or the like, and also on covers which, owing to their structure do not safeguard against falling objects i.e. bolts, tools, welding electrodes, etc.

34. Noise of Work:

- Full account must be taken of the requirements of the relevant noise at work regulations.
- Certain areas on site require the use of hearing protection and NOMINATED SUBCONTRACTOR are responsible for ensuring that their own staffs are properly equipped to enter and work in these areas.
- Where NOMINATED SUBCONTRACTOR are intending to use equipment or machinery which can give rise to noise levels in excess of 85 d B(A) then appropriate measures must be taken to comply with regulations and suitable control by use of elimination, substitution, engineering control, administrative control and or the use of personal protective equipment.
- NOMINATED SUBCONTRACTOR has to carry out monthly noise survey at site and various locations at noise generating operations and take adequate measures to control the same, if it is above specified limits.

35. Inspection of Tanks/Confined spaces:

- Before any inspection is carried out within tanks or pipes or in narrow spaces, etc.an inspection permission, permit to work procedure shall apply, must be issued by the site or project manager of the responsible NOMINATED SUBCONTRACTOR and approved by the site management prior to commencement of the work.
- The NOMINATED SUBCONTRACTOR shall ensure that all practical steps have been taken to remove dust, fumes, gases or other impurities and danger which may be present to any confined space or trench or excavation is certified by the responsible competent person to be safe and fit for the entry of workers.
- In confined spaces the oxygen level should be checked and also gas test should be carried out.
- For carrying out work in confined space work permit should be obtained and required safety precautions to be taken.

36. Transport Equipment and lifting Gear:

- The user of the equipment bears the responsibility for the sufficient carrying capacity, suitability and safety of the transport equipment and lifting gear, for the proper handling of his own equipment and of equipment provided by others, and for all the necessary protective facilities.
- The user must satisfy himself that the equipment has been tested and inspected, and is properly maintained.
- The NOMINATED SUBCONTRACTOR shall continuously monitor the equipment, including load carrying facilities and means of attachment, and if any deficiencies arise or are suspected, the NOMINATED SUBCONTRACTOR must immediately ensure that the device in question is taken out of use.
- No item of lifting equipment shall be used on the site unless a test certificate in accordance with statutory requirements has been supplied by the NOMINATED SUBCONTRACTOR for review by site management at site.

DRAFT

- The NOMINATED SUBCONTRACTOR is responsible for ensuring that test certificates are maintained and the prescribed register is duly completed during the contract period and renewed by current legislation to maintain validity.
- Prior to starting any erection work ,lifting work NOMINATED SUBCONTRACTOR must ensure all lifting tools, tackles, equipment having a valid test certificates as per the applicable statutory regulations.
- All safety devices are to be inspected and maintained in good working order. Records of such inspections and maintenance are to be kept by the responsible member.

37. Radioactive communications, Telephones:

- Radiotelephone communications on site shall be subject to the appropriate laws and regulations. Anyone using radiotelephone equipment shall himself ensure that he has the appropriate permits. Equipment for which no permit has been issued must not be operated on site. After permit has been received, the site management shall be informed of the call signs, the number of devices and the operating frequencies.

38. Workplaces and facilities belonging to other companies:

- Unauthorized persons are not permitted to enter the erection areas, storage areas or accommodation areas of others or to step on the construction scaffolding or erection scaffolding of other contractors.
- The representative of the authorities, of the GENERAL CONTRACTOR and of the site management are exempted from this stipulation.
- If entering erection areas belonging to other contractors or facilities which are in operation and belong to other contractors, it is unavoidable (e.g. in order to reach one's own erection area), the agreement of the erection manager of the company in question or of the site manager must be previously obtained.
- Erection equipment belonging to other contractors such as crane facilities, erections masts, lifts, bulldozers, vehicles, etc., must in no event be used without prior agreement of Project manager. The same applies to the crane facilities, other lifting gear, lifts, machinery, devices and the like, which are part of the final equipment.
- Places of work are to be left in a safe and tidy condition at the end of each work period.

39. Procedure for obtaining working permission;

- Pressurized, hot media, live and electrical systems are connected successively during the course of commissioning .Work performed on these facilities or on other facilities indicated as falling within the range of applicability of the procedure for obtaining a work permit are subject to special approval or to individual instructions issued by the site management.
- Work falling within the range of applicability of the procedure for obtaining working permission must not be performed without special permission of the GENERAL CONTRACTOR's site management, as demonstrated, for example, by a valid certificate of working permission.

40. Unauthorized operating of switchgear and valves:

- The unauthorized operation of switchgear and valves of commissioned plant components must be avoided under all circumstances, since the sudden starting can lead to severe accidents or damage to the plant.

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41. Alcohol and drugs:

- It is strictly prohibited to consume or be under the influence of alcohol or non prescribed drugs on site.
- No alcohol is permitted on site, and anyone deemed to be under the influence of alcohol shall be required to leave the site immediately.
- Contravention will lead to disciplinary or legal action.

42. Mandatory requirement:

- No workmen below 18 years and above fifty eight years of age shall be engaged for the job.
- No domestic LPG cylinder is allowed for hot work such as gas cutting etc.
- All statutory regulations of the state should be strictly followed including labor license, migratory labour license, WCI, ESI, P.F, Muster roll, minimum wages, required licenses, insurance policy and approvals from local ,legal or government offices should be taken and maintain a copy.
- All documents in accordance with applicable local laws and statutory regulations.
- All powered cranes , hoists, aerial platforms and scissor lifts, lifting equipments, tools &tackles if any must be certified as safe by a third party, or be a person or persons holding a certification recognized by local or national legislation and have a competent driver.

43. No smoking:

- Smoking on site is restricted except in designated place.

44. Non-compliance of safety procedure:

- The non-compliance of any safety rule, regulations and applicable statutory requirements observed will call for a in disciplinary action as per GENERAL CONTRACTOR's Non-compliance safety procedure.

46. Environment protection:

- The environment control programme shall be implemented on site to address the environmental aspect of the project activities, necessary for the execution of the project.
- NOMINATED SUBCONTRACTOR having the responsibility shall ensure the relevant activities and fully cognizant of, and abide the environment management programme.
- NOMINATED SUBCONTRACTOR, each contractor shall be responsible for ensuring full compliance of all Environmental protection obligations with at the construction site are defined in stipulations of the applicable laws and ordinances in the provision of his scope of supply and/or services. Any and all requirements imposed by licensing and supervisory authorities for the construction and erection time period (e.g. low-noise construction equipment) shall also be met.
- NOMINATED SUBCONTRACTOR, each contractor shall be responsible for ensuring full compliance with GENERAL CONTRACTOR's Environment protection programme requirements.
- Compliance with the provision of the above mentioned shall not relive any contractor of his environmental responsibilities or of any obligation imposed on him by any act, regulation, order or by-law made by the competent authority having the force of law in India that may be applicable to the project.

DRAFT

- Site ambient air monitoring should done on monthly basis by MPCB approved agency and report to be submitted to client.

47. Data Protection, information security:

- All contractors represented at site are obligated to handle the data and information made accessible to them within the framework of order execution in accordance with the specified protection clause.
- NOMINATED SUBCONTRACTOR, each individual contractor is obligated to ensure compliance with data secrecy requirements (in particular in the processing of personal – specific data).
- Unauthorized persons shall be, strictly prohibited obtaining access to data and information.

48. Recordkeeping:

- Reports as stipulated by the site management in the form of daily ,weekly, monthly or other records of the employment of personnel ,progress of work, use of equipment, deliveries of material etc., must be done in time by NOMINATED SUBCONTRACTOR, each contractor. If necessary the site management will issue forms. If is NOMINATED SUBCONTRACTOR's responsibility to make the necessary copies to be filled in and submit the reports at the times stipulated by the site management.

49. Permit to work:

- NOMINATED SUBCONTRACTOR shall issue safety work permit system, which is usually considered for job or task that is not normal or routine. Some of the tasks for which permit shall require Hot work(Welding ,Gas cutting etc.),Excavation job, Confined space entry, Lifting operations, Work in electrical sub-station or on electrical circuits/equipment, work at height (above 3 m)

50. Signage's:

- NOMINATED SUBCONTRACTOR should arrange necessary signage's, Safety posters, safety slogan, safety instruction boards shall be displayed at site on proper manner to create safety awareness amongst staff and workers.
- Safety Sign boards: Various types of message of cautioning, attention, notice, etc. should be displayed at the appropriate places for learning/awareness of the workmen while working at site.
- Suitable warning signs to be displayed warning against potential hazards.

51. Working personnel:

- Each working personnel on site shall support all measures for accident prevention.
- In particular they are obliged to follow instructions aimed at accident prevention and reject instructions which are prejudicial to safety.
- The personnel working on site may use facilities for the purpose they are designed for unless otherwise advised by the site management.
- If the personnel observe that a facility within a NOMINATED SUBCONTRACTOR's area of competence is not faultless from the safety point of view, they shall immediately eliminate this deficiency. Otherwise they must provisionally safeguard the danger spot and immediately report the deficiency to their superiors.

DRAFT

- In the event of accidents or of a general danger and distress, everyone has the duty to provide help to the extent that is necessary and can, in circumstances ,be expected from him, in particular without running any considerable personal risk nor neglecting other important duties. If a site Rescue Team is formed, NOMINATED SUBCONTRACTOR /contractors are obliged to participate in co-operation with the site management.

52. Personnel Protective Equipment:

- The NOMINATED SUBCONTRACTOR shall provide all necessary approved personal protective equipment to his employees appropriate to the hazards for which protection is required.
- The following PPE`s are generally used on sites. However special requirement may be required for certain operations depending upon the risk.

1. Mandatory requirements	
Safety Helmet	Should be used by all persons working on site.
Safety Shoes	Should be used by all persons working on site.
Reflective Jacket	Should be used by all persons working on site.
2. Special requirements	
Eye protective safety goggles/face shield/Welding face shield etc.	For gas cutting, welding, grinding, chipping/chiseling/wood cutting/hacking/grinding/demolition/breaking stone/handling chemical/drilling/painting etc.
Hand protective gloves: leather hand gloves	For steel reinforcement, material handling, unloading of reinforcement bars, bending& cutting of bars ,glass handling, tin sheet handling, or handling of sharp objects etc.
Rubber hand gloves	For concreting, removing water, earth
Appropriate protective equipment.	For handling alkalis, acid, or other similar corrosive substances
Gum boots	For concreting, working in mud and water, chemicals etc.
Safety Harness/life lines/fall arrestor	For work involving height to prevent fall of person from height.
Apron/safety clothing	For welding, gas cutting, handling chemicals
Ear plugs/muffs	Places, equipment or operation where noise level is high.
Water proof rain coat	For work in rain of similar wet condition.

- Any person observed violating these basic safety norms would be eligible for strict action.
- The NOMINATED SUBCONTRACTOR shall have sufficient stock of various PPE`s in his store room to avoid any shortage of supply and these should be maintained in good condition.
- Damage or defective PPE`s should be replaced with good ones.

DRAFT

- The NOMINATED SUBCONTRACTOR shall give training, informing the importance of PPE`s and demonstrating how to use, to his employees.
- The NOMINATED SUBCONTRACTOR's shall give status of various PPE`s in prescribed format should be submitted to GENERAL CONTRACTOR's office every 15 days.
- Follow GENERAL CONTRACTOR's standard helmet colour coding.

53. Machine guarding:

- The NOMINATED SUBCONTRACTOR shall ensure that dangerous Moving parts of all equipment/portable or fixed machines /power tools etc.in use are adequately covered or guarded.

54. Machine, tools, Equipments Entry and Inspection:

- All equipment /machine sets like welding /gas cutting machines ,grinders, drill machines, cutters, bar cutting &bending machines, vibrators, grinders, and any other machine which not mentioned above should be inspected from electrical Engineer and certified as safe to be used on site.
- The inspected machine should have inspection label with signature of the inspected personnel.
- No further addition or modification should be allowed once machine, tools, equipment inspected. Any alteration made should bring into notice and machine, equipments, tools should be again inspected.

55. Maintenance regimes:

- All the Plant, machinery, equipment, lifting equipment, cranes, passenger hoist, material hoist and any other machine used at site shall be controlled, periodically checked & regular maintenance shall be carried including preventive maintenance ,plant maintenance & site complaint/repair) ,plant maintenance (vehicle and equipment repair/servicing) and maintained in good conditions and records of maintenance to be kept.
- Maintenance log sheets/records of maintenance of all plant, equipment, machineries, vehicles, cranes, lifting and hoisting equipment's, etc. should be maintained at worksite as per recommendations of manufacturer service manuals.

56. Electrical hazards:

- All electrical work at site shall be performed by approved qualified electricians.
- Persons carrying out the electrical works shall be trained, knowledge and possess valid license and a copy of the license to be submitted to GENERAL CONTRACTOR's office.
- The electricians shall have required tools of standard make, sound construction and good quality for carrying out work safely. Electricians shall use appropriate tested insulating Rubber hand gloves, shock proof rubber shoes /boots of correct voltage grade while working on live panels and rubber mats are places at electrical panels.
- Before commencement of any building or other construction, the NOMINATED SUBCONTRACTOR shall take adequate measures to prevent any worker from coming into physical contact with any electrical equipment or apparatus, machines, or live electrical circuit which may cause electrical hazard.

DRAFT

- The NOMINATED SUBCONTRACTOR shall display and maintain suitable warning signs at conspicuous places in Hindi and in a local language understood by the majority of the workers.
- The NOMINATED SUBCONTRACTOR shall ensure that all electrical installations/work must be done in accordance with the valid statutory requirements.
- In workplaces where exact location of underground electric power line is not known ,the workers using jack hammers, crow bars or other hand-tools which may come in contact with alive electrical line ,shall be provided with insulated protective gloves and foot-wear.
- Section applies to buried lines: In the case of open laying, cables must be protected against unintentional damages, laid free and maintained, i.e, they must in no case be covered, obstructed or encumbered.
- The NOMINATED SUBCONTRACTOR shall install and continuously monitor the electric circuits, distribution boards, connecting cables and consuming devices with the help of qualified electricians.
- The NOMINATED SUBCONTRACTOR shall ensure that all electrical appliances and current carrying equipment used are made of sound material and are properly and adequately earthed.
- Earthing pit provided at site & routing should be done as per electrical code of practices.
- All electrical switch boards/temporary electrical installations to have ELCB`S, MCB`S to act in case of any improper connections/overloading/earth leakage, etc.
- All electrical circuits/electric distribution boards/sub-distribution boards, equipment used by NOMINATED SUBCONTRACTOR will have proper double earthing connections will be connected through an ELCB (rating of the ELCB should be 30 mA earth leakage) from earth pit.
- Proper electrical fixtures should be used for power supply. Wires inserted into socket are not allowed.
- Proper rating cables (3/4 core industrial cables) should be used. All cables supported above the ground, routed through height in a systematic manner. All cable supports non-conductive.
- All electrical installation shall be properly earthed to ground and protected from rainwater. Proper covering/shade shall be provided to protect it from rain water. Periodic inspection of the installation should be carried out and documented and a copy to be submitted to GENERAL CONTRACTOR's office every month.
- Earth resistance periodically checked and documented.
- All lines de-energized (isolated & tagged) before carrying out any connection /repair work "LOCK &TAG OUT" SYSTEM TO BE USED .Caution shall also be displayed to prevent any accidental start of power supply.
- No open cable connections, use of standard connections for joining the electric cable.
- All electrical equipment and appliances shall perform to applicable BIS standard /Indian electrical rules.
- Metal ladders must not be used where there is any possibility of the ladder coming into contact with electrical lines.
- Shock treatment charts to be displayed at all important locations as per Indian Electricity rules.

DRAFT

- Cables routed /underground shall be clearly marked /display cable routes and cable marker, so as people will get alert about underground cables and prevent any damage.

57. Lighting arrangement:

- Adequate area lighting and job task lighting must be provided to enable safe access and egress from every place on a site where persons are liable to work.
- Adequate area lighting shall be provided from security entrance gate to site workplace and labour camp access for safe movement.
- The illumination and electric tools used shall meet the statutory regulations or other equivalent regulations.
- Provision of fire extinguisher & dry sand buckets shall be made at all main electrical distribution boards.
- The NOMINATED SUBCONTRACTOR shall install additional lights at places/location or areas as specified by the consultant / GENERAL CONTRACTOR from safety point of view.

58. NOMINATED SUBCONTRACTOR and sub-contractors Tools and Equipment:

- All contractors and sub-contractors tools and equipment must comply with statutory regulations and approved codes and practices.

59. Hand Tools &Electrical driven portable tools:

- It is NOMINATED SUBCONTRACTOR's responsibility to ensure that the tools he works with are suitable for the job and in a safe condition prior to work commencement.
- All power tools shall be inspected by electrical Engineer before being used at site.
- Right tools shall be used for the right job.
- Electrical portable hand tools should be 110V type with suitable step down transformer for converting 220V to 110 V.
- If 220 V type used necessary protection devices like ELCB/RCCB &proper earthing should be provided and maintained.
- For Cables connections of the power tools shall use appropriate pin plugs.
- Makeshift connections are prohibited.
- In all cases, with the exception of double insulated tools, the metal work of the tools must be effectively earthed; also any flexible metallic cable coverings must be earthen.
- Defective tools shall be discarded.(Cracked, broken edge, mushroom head ,worn out, loose handle etc.)
- Periodic inspection of tools shall be carried out.
- Always maintain hand tools in good condition.
- Hand tools shall be carried in tool box or pouch.
- Guards provided along with tool should be in place.
- As far as possible use battery operated & cordless) power tools.
- Provide proper earthing to all power operated tools, equipment.

Compressed Air tools:

- NOMINATED SUBCONTRACTOR and sub-contractors must obtain permission to use any air supply on confirming party site.
- NOMINATED SUBCONTRACTOR and sub-contractors must also provide suitable noise suppression for pneumatic hammers, drills, etc.

DRAFT

Tool operator:

- Only operators who are fully trained in the use of such tools shall be allowed to operate them on site.
- All operators of any explosive – powered tool shall be qualified and licensed in accordance with statutory requirements. A warning sign shall be displayed at each place where the tool is being used.

60. Operations Adjacent to overhead lines:

Work near the overhead lines as per the Indian Electricity rule.

- Work activity adjacent to overhead lines shall not be initiated until survey has been made to ascertain the safe clearance from energized lines.
- Overhead transmission and distribution lines shall be carried on towers and poles that provide safe clearances over roadways and structures.
- Clearances shall be adequate from the movement of vehicles and for the operation of construction equipment.
- Any overhead wire shall be considered energized unless the electrical line certificates that it is not energized.
- Work activity that could affect or be affected by overhead lines shall not be initiated until coordinated with the appropriate utility officials.
- Materials and equipment's shall not be stored under energized lines, or near energized equipment.
- Stop work if you find something that may be a service found unsafe, report it straight away.
- Where service is damaged, signs, barricades etc. must be displayed.
- Report any damaged services as soon as possible. Don't work commence until it is safe.
- Operations adjacent to overhead lines are prohibited unless the minimum Clearances for live –line bare as mentioned in electricity rule and adequate safety precautionary measures taken.

61. Lightening protection:

- Hazards occurring to persons or materials' which will increase lightning strike risks, e.g. cranes, masts, etc., must be checked regularly.

62. Material storage:

- All building and construction material shall be stacked properly on markup area.
- The material should be stacked in proper position at required height and should be in stable condition.
- Follow ISI code for stacking various kinds of materials.
- Material or equipment is not stored or placed so close to any edge of a floor platform as to endanger the safety of persons below or working in the vicinity.
- Flammable and combustible materials shall be stored in closed room, well ventilated, protected from heat, rainwater and equipped with fire extinguishers.

DRAFT

Exit and Aisle way requirements:

- Material is not allowed to be stored in or near any interior or exterior exit doors, aisle ways, emergency exits and stairwells.
- Material storage at higher floor shall be done properly. No material storage near open floor edges/floor openings, etc. Securely storage of material within floor area.

63. Excavation & Trenching:

- Prior to start an excavation, underground services soil survey shall be carried out, located the services and protected from damage or displacement. NOMINATED SUBCONTRACTOR shall obtain excavation work permit from GENERAL CONTRACTOR.
- All vehicles and earth moving equipment used on site shall be in good working condition and are maintained in good state and properly used.
- NOMINATED SUBCONTRACTOR shall ensure all the earthmoving equipment should have due certificates such as insurance, operator license, PUC certificates before commencing the activities within the premises and a copy of certificate shall be submitted to GENERAL CONTRACTOR's for record purpose.
- Machine like excavator, JCB, Poclain etc. used shall be inspected to ensure it is safe to use on site e.g. Brake, horn, swing movement, lights, no leakage of oil etc.
- Only trained and experienced operator shall operate the machine.
- Helper shall be available with operator to guide him.
- Only operator shall sit in the machine and no other person is allowed to site in the machine while in operation.
- The swing radius area of the machine should be cordon off to prevent persons coming in the danger zone of the machine.
- Avoid overloading of material during transportation. Vehicle moving out of site into public access road should be covered properly; no loose material shall be transported. And ensure adequate tyre cleaning/washing measures & dispose water should routed to normal storm water channel after sludge separation.
- Traffic control person /signal man should be employed to guide the vehicle movement safely and to avoid traffic jam and congestion.
- The excavated material, soil shall be deposited 1.5 meter away from the edge of the excavated pit/trench or half of the depth of trench whichever is more. Excess loose earth shall be disposed off from the site regularly. Loose earth kept on site for backfilling shall be leveled and stacked properly.
- All trenches and excavations wall surface shall be protected by means of shoring, step cutting, slope cutting of ground, shore piling, or any other safe method to avoid the danger of slides and to prevent of event collapse of the wall while people are working in the pit.
- The NOMINATED SUBCONTRACTOR shall ensure the stability and safety of the excavation, adjacent structures, services, and the works.
- Protective systems shall have the capacity to retain without failure all direct and indirect loads that could be expected to be applied to the system.
- Submersible pumps shall be available at site to remove water collected in the pit.
- Proper strong and safe access shall be provided in the pit for entry and exit of workmen.
- Any loose soil or stone on the sides of the pit shall be removed before allowing entry of people.

DRAFT

- The excavated pits should be hard barricaded by rigid material at a distance 1 m from the edge to prevent personal, vehicles, and equipment from falling into excavations.
- The area of excavated pit shall be adequately illuminated after day's work and danger boards displayed.
- Competent person/excavation engineer shall inspect the excavation, the adjacent areas, adjacent structure and protective measures/systems daily, as needed throughout the work shifts, and after rain storm or any other hazard increasing occurrence.
- All trenches shall supply with access ladder. Ladder shall be extended from bottom of the trench to at least 1 m above the surface of the ground.

64. Blasting:

The NOMINATED SUBCONTRACTOR shall ensure the following safety precautions for blasting operation:

- License NOMINATED SUBCONTRACTOR should be appointed for blasting operations.
- All blasting operations shall be carried out on the basis of procedures approved by inspector of explosives .All works in this connection shall be carried out as per IS codes of practice.
- The NOMINATED SUBCONTRACTOR shall appoint a person suitably qualified experienced in the use of explosives that who will have total control over all blasting operations on site.
- All measures and precautions shall be taken for use, handling, storing or transportation of explosives to prevent personal injuries and loss of property.
- All laws, in force, shall be complied with.
- NOMINATED SUBCONTRACTOR shall comply with the regulations & procedures in storage, transport, handling, use of the explosive material according to the guideline or rules framed under Indian explosive act.
- The NOMINATED SUBCONTRACTOR shall comply with the local regulations under district collector jurisdiction.

65. Piling Work:

The NOMINATED SUBCONTRACTOR shall ensure

- Pile driving equipment should be tested for sound construction, good design are properly maintained.
- A responsible person having adequate experience and knowledge for inspecting pile driving equipment shall inspect such equipment before taking it into use and shall take all appropriate measures as required for the safety of construction workers before commencing piling work by such equipment.
- Lifting equipments used shall be tested.
- A pile driver is firmly supported on a heavy timber sill, concrete bed or other secured foundation.
- In case a pile driver is required to be erected in dangerous proximity to an electrical conductor all necessary precautions are taken to ensure safety.

DRAFT

66. Height work:

Scaffolding

- Work scaffolding and guard scaffolding must be erected on the site by competent qualified scafolder s in accordance with statutory requirements and other stipulations applicable.
- For scaffolding which bears substantial loads, the NOMINATED SUBCONTRACTOR must hold available static calculation documents which can be checked, authorizations and test certificates issued but the official civil engineering or construction inspection authorities and stress analysis.
- The NOMINATED SUBCONTRACTOR using the scaffolding will be responsible for its proper use.
- The user of the scaffolding shall satisfy himself whether it is suitable for the envisaged utilization. This should be particularly observed when the scaffolding is also used by other persons of companies.
- Ensure Scaffold erection is carried out in accordance with standard scaffold erection practices & the scheme shall be as per approval of site In-charge approved design.
- A sketch of the scaffolding proposed to be used shall be prepared and approval of the Site engineer obtained prior to start of erection of scaffolding.
- Scaffolding or staging more than 3.5. m above the ground or floor ,swung or suspended from an overhead support or erected with stationary support shall have standard guard rail properly attached ,bolted, braced or otherwise secured at least 1 m. High above the floor or platform of such scaffolding or staging. The guard rail shall extend along the entire exposed length of the scaffolding with only such opening as may be necessary for the delivery of materials. Such scaffolding or staging shall be so fastened as to prevent it from swaying from the building or structure. Scaffolding and ladder confirm to relevant IS specification.
- No scaffolding, ladder, working platform, gangway runs, etc. shall exist within 3 meters of any insulated electric wire. Whenever electric power and lighting cables are required to run through ,(pass on) the scaffolding or electrical equipments are used, such scaffolding structures shall have minimum two earth connections with earth continuity confirming to IS code of practice.
- NOMINATED SUBCONTRACTOR shall engage competent scaffolding person/ qualified scaffolding supervisor and trained erector for the erection, maintenance and dismantling of scaffoldings.
- Scaffoldings must be inspected by appointed scaffolding supervisor periodically and inspection records maintained.
- Scaffolding after taken into service must be periodic visual inspection & maintenance of scaffolding members and any defects to be rectified immediately to ensure that it is safe for use.
- After completion of repair work/erection work of scaffold, submit the report of stability and design load carrying capacity to GENERAL CONTRACTOR's.
- Prior to allowing workers access to the scaffold the scaffold supervisor shall inspect the scaffold to ensure that it is safe for use.
- Before start of work job specific tool box to be given.
- All loose materials on scaffolding to be removed immediately.

DRAFT

- Wooden ladders /Godhies shall not be used for working at height .properly designed scaffolding and platforms of steel with suitable guard railing shall be used.
- Instead of wooden Ghoda use Aluminum thristle Ladder/mobile scaffold for powerless connections.
- Only workers capable of scaling heights should be deputed to do the jobs.
- Major scaffolding used onsite must be metallic. The bamboo/wooden scaffold should be discouraged with consideration of safety. Materials used in the scaffolding shall be of sound material, good quality, sound construction and free from any defects.
- All scaffolding used on site must be design &erected to support intended loading on it.
- Scaffolding erected shall be properly braced, interlocked and supported to ensure it is stable.
- Scaffolding shall be erected on firm ground with base plate. Strong and sturdy working platform with guardrails shall be provided. The working platform shall be properly tied with the frame on both ends. All working platform –strong, sturdy, free of defects and should have adequate fall protection guardrails, midrails &toe boards provided.
- For tall scaffold; there should be suitable means of climbing.
- Proper means of access shall be provided to reach the working place; person shall not climb scaffolding to reach at work place.
- Choose the right ladder length. Ladders used to be checked for missing rungs, cracks, broken legs, antiskid bush etc. and the fitness of the ladder shall be certified by the NOMINATED SUBCONTRACTOR's engineer at site. Standard ladders shall be used.
- Men working at heights should be provided with tool kit pockets for keeping hand tools for easily carrying at height.
- Do not throw materials or hand tools from ground or drop them from height.
- While working at height make safe arrangement to prevent materials falling on ground.
- Work at more than one elevation at the same segment to be restricted.
- While height work is in progress, the area below shall be cordon off.
- Fall protection must be worn when employees are working near open edges of structures where workers are likely to fall.
- The walkway jallies used for working platform, it shall be adequate strength to take the load of men and materials and shall not collapse. The work-platforms shall be rigidly tied at both ends to prevent sliding and slippage.
- Bamboo scaffolding: In general, it should be avoided as far as possible. It should not be used in the unit /off-site areas and where hot work is to be done.

67. Safety screen/ net:

- Provision of Safety net at podium level above 6 m to protect persons from falling & falling objects.
- Provide safety net below working level and de-shuttering level till completion of de-shuttering at higher floors to protect falling object and fall of persons.
- Provision of vertical safety screen (as per below photos) surrounding periphery of working level and 2 level below of working level for arresting the wind/screening of people with vertigo and to provide fall protection.

DRAFT





- Safety net at other unknown place like Public road, podium level protection, access way, Site entrance, labour camp access pathway, etc. shall be provided.
- Provide viable covering net surrounding building to prevent fall of material from intermediate level by jumping off.
- NOMINATED SUBCONTRACTOR has to follow standard work practice of Safety net installation with proper supporting arrangement as per the standards.

68. Mobile scaffold:

- Mobile scaffold needs to be designed and erected such that it should be stable and safe to work.
- Free standing scaffolding towers used externally must not be higher to the top platform level than four times the minimum base dimension, unless secured to a permanent structure. The height of the platform may rise to 3.5 times the minimum base dimension. For mobile working scaffolds, wheels must be locked when towers are in use. No person is permitted to remain on any tower platform whilst the tower is being moved.

69. Suspended scaffolds (Cradles/ Gondolas):

- Suspended scaffold used for must be capable of supporting the slung or suspended platform. The suspension arrangements must be strong, enough and the platform itself should be sufficiently robust to carry the intended load of people and materials with guard sides or rails to prevent them from falling out.
- Suitable overhead protection for the worker working need to be provided against falling object from overhead/height.
- Daily before using concerned engineer or designated competent supervisor should be check all components, weights, wire ropes, anchoring arrangements, electrical connection, emergency devices etc.to ensure effective functioning of suspended platform. The electrical equipment should be having double earthing which need to be check periodically and the worker will have physical inspection before starting the work.
- When the position of the suspended platform shifted from one place to another it should be checked by a competent person for ensuring the proper re-assembling.
- Suspended platform before taken into service should be tested by government authorized competent agency.

70. Safety barrication:

- Guardrails are to be provided at all working places and other locations where persons or material could fall above 2 m heights as per IS 4912.

DRAFT




- Barricading of open edges &cut-outs as per IS 4912
- Temporary hand rails for staircase.

71. Safety harness:

- All workers working at height over 2 m from ground level shall wear full body safety harness with double lanyard ; the other end of the lifeline shall be hooked /anchored to a firm support at all places of work.
- Provision of life lines (Horizontal and vertical).Proper anchoring points.
- If the form support is not available then NOMINATED SUBCONTRACTOR's Engineer shall provide life line for hooking /anchoring the safety harness.
- Full body safety harness with fall arrestor devices must be used while working in lift shaft, erecting scaffolding, external painting, working in cradle.

72. Floor openings:

- Every opening in the floor of a building or in a working platform shall provide with suitable means to prevent fall of persons or materials by providing suitable covering or fencing or railing as per IS 4912.
- Unprotected openings in floors & edges shall be protected with securely covers and /or have guardrails to protect employees
- All Holes, shafts and edges from or through which persons could fall must be adequately protected by covers or barriers so as prevent falls of persons and materials.



73. Roof work:

- Roof ladders for working on fragile roofs:

To work on AC roof requires special arrangements and knowledge of working. A special type of portable ladder called "CRAWLER –LADDER" IS DEVELOPED TO WORK ON ROOF TOP. Proper method of placing roof ladders and crawling boards.

DRAFT

- The person working on AC sheet roof must use safety belt with lifeline attached to independent anchoring arrangement.
- Roof working must be properly supervised.

74. Overhead protection:

- The NOMINATED SUBCONTRACTOR shall ensure that overhead protection is provided along the periphery of every building under construction.
- The NOMINATED SUBCONTRACTOR shall ensure that any area which is exposed to risk of falling materials, articles/objects is or cordoned off or otherwise suitably guarded from inadvertent entry of person other than construction workers.
- No work may be carried out above the heads of people or over gangways or roads, until all precautions have been taken to ensure the safety of the person below. Each specific site of overhead working shall require consent from the safety officer. This shall be given after satisfactory inspection.

75. Welding, Gas cutting work:

The NOMINATED SUBCONTRACTOR shall ensure the following for welding, gas cutting and grinding work at site;

- Proper equipment check before use. Electrical engineer of the NOMINATED SUBCONTRACTOR shall electrically certify the all machines prior to its use on site.
- Welding sets shall be properly earthen through insulated cable to the nearest earth pit.
- Welding machine shall have ON/OFF switch on the machine and in working condition.
- Each welding and gas cutting set provision of of fire extinguishers.
- Welding cable should be free from any joints and insulation should not be damaged at any place. Welding cable shall have standard 3 pin plug with earthing point.
- Earthing cable shall be free from joints, properly insulated and sufficiently long for the earthing of the job. Usage of rods for earthing of the job is not permitted.
- Only qualified, trained and experienced welder & gas cutters should perform the welding and gas cutting work. Helper is not allowed to operate welding machine.
- Ensure that all necessary PPE`s will be used by the workmen during operation like hand protective gloves, shock proof safety shoes, eye protection welding face shields, gas cutting goggles etc. and safety harness if working height.
- Welding & gas cutting operations should be performed away from flammable materials. if the object to be welded cannot be moved to a safe location, if this cannot be accomplished completely, the welding zone should be enclosed in fire proof blankets or other protective shields when materials in nearby areas can be affected by welding arcs, flames , sparks , spatter ,slag or heat. All movable hazardous materials should be moved to a safe location
- Regulator, hose, torch, welding cables and joints etc. should be in good condition without damage, cuts or open joints.
- Gas cutting set & cylinder shall always be handled in trolley.
- Gas cylinder & gas cutting torch shall be fitted with suitable flash back arrestor.
- Gas cylinder shall be fitted with pressure gauze, which should be in good working condition.
- Gas cylinder should always be kept in upright manner.

DRAFT

- After the completion of work close the valve of the gas cylinder and discharge the gas from the hose pipe.
- At the end of shift or after completion of work gas cylinders shall not be allowed to keep/left in working place.

76. Material handling/ erection work:

- The NOMINATED SUBCONTRACTOR shall ensure that all lifting appliances and plant or gear used in anchoring of fixing of such appliances are of sound construction, sound material and of adequate strength and are always maintained in good working condition.
- Test and periodical examination of lifting appliances: The NOMINATED SUBCONTRACTOR shall ensure that all lifting appliances tools &tackles including all parts and gears thereof, whether fixed or movable are tested and examined by government approved competent person before taken into use for the first time and after it has undergone any alterations or repairs liable to affect its strength or stability or after erection on construction site. All lifting appliances shall be thoroughly examined by a competent person, at least once in twelve months. The competent person, after making such examination shall form the opinion in writing that the lifting appliances are safe for use for the intended construction work.
- Loading of lifting appliances and lifting gears: The NOMINATED SUBCONTRACTOR shall ensure that no lifting appliance, lifting gear or wire rope is used in an unsafe way and in such a manner as to involve risk to the life of workers and that they are not loaded beyond their safe working load.
- All lifting appliances and lifting gears, tools &tackles etc. should have valid test certificate.
- Lifting equipment or any other equipment shall not operate adjacent to energized transmission lines& near overhead lines. The lines shall be de-energized when operating equipment or moving materials near such energized equipment.
- Operators must have all required licenses and certificates for the equipment being operated.
- When lifting appliances/equipment used, an experienced banks man shall be provided for safe operation and communication to crane operator.
- Hook or load riding shall not be permitted except in accordance with statutory authority nor shall passengers ride on material hoist, bulldozers, front end loaders, forklifts or similar plant not intended for the carrying of personnel.
- Employees cannot work in areas directly beneath a suspended load.
- Cranes cannot be operated above the roof of an occupied building or area.
- Cranes must be equipped with an audible warning device.
- Mobile crane cannot be operated near overhead electrical power lines, maintain safe distance as per IE rules.
- Any contractor using Lifting appliances, Tower crane, passenger hoist, material hoist, lifting tools, tackles equipment etc. without valid certificate shall not be permitted to use on site before commencement of work conduct third party inspection and Submit valid test certificate.

Erection:-

- Prior to the erection of work plan the activity and prepare a erection plan or scheme. After the erection plan task hazard analysis should be carried out as per the sequence of operation

DRAFT

to know the hazards at each operation and to arrive at suitable safety precaution to overcome the possible hazard for averting any accident or incident.

- A copy of the task hazard analysis has to be given to the erection contractor for carrying out the erection operation as stated in the document for safe and easy erection operation.
- The lifting equipment or machinery (Crane, hydra, lifting accessories, tools and tackles etc.) Which is being used for erection operation shall be tested one and of required capacity. A copy of test certificate of all lifting equipment should be obtained from the NOMINATED SUBCONTRACTOR s duly verified before putting into use.
- A Work permit for the erection job should be issued to the NOMINATED SUBCONTRACTOR prior to commencement of work.
- Engage tested & certified crane & lifting gears. Inspect the lifting gears every day for its defects.
- A proper and safe access to be provided for the erection person to carry out the erection work.
- To prevent unauthorized persons entry area should be cordon off or barricaded to prevent unauthorized entry.
- Designate trained & experienced personnel for signaling.
- Use PPE`s like, safety helmet, safety shoes, hand gloves, safety harness, life line, fall arrestor etc.
- Use proper ladder for safe climbing /descending &hold it to prevent sliding.
- Stop lifting when wind pressure is high.
- Use proper guide rope for control the swing.
- Check clearance of nearby facilities, if required obtain Permit to work
- Cautionary board "ERECTION WORK IN PROGRESS", "DANGER KEEP AWAY" should be displayed at site.
- Work under the vigilant supervision of Engineer or his designated supervisor who is competent in erection operations.
- The person involved in this activity should be trained, knowledge of erection work. Before starting daily activity give detail tool box talk and issued all necessary personal protective equipment.
- The person working at height should wear full body harness and its rope to be tied /hooked/fastens to a strong and firm support.
- Hand tools should be carried in tool box or pouch.
- No materials shall be thrown from ground or dropped from height. The material can lower or lifted with the help of rope.
- Only authorized persons to give signals during lifting or lowering, shifting of structures.
- The area of erection should be cleared off other activities.
- No person should walk on beams or steel structures.
- Erection work should not be allowed after day light.
- Monitoring of implementation of all safety precautions during erection as stated in the task hazard analysis should be carried out.

DRAFT

77. Chemicals and inflammable materials (oils, diesel, petrol etc.)

- Chemical brought to site must be labeled for its contents, its hazards and the Material safety Data Sheet (MSDS) be provided to Project Manager GENERAL CONTRACTOR's. Common household chemicals are exempted from this requirement.
- The stores for the chemicals should be made in an area which is secluded from any other stores/godowns/site offices to avoid spread of the fire in case of emergencies. There should be enough ventilation and stored as per MSDS and manufacturer's instructions.
- Adequate firefighting arrangements to be maintained near the stores where these chemicals will be stored.
- Where petrol, diesel or gas powered equipment is used on site with, fuel therefore shall be stored only in satisfactory metal safety containers in accordance with statutory requirements.
- The stores should have an hard flooring done to avoid any seepage of the spill over from the containers, in the bare ground earth. A bund should be made to contain the spillage.
- Storage of empty and full compressed gas cylinders shall involve proper securing of every gas cylinder in upright position, separate storage of full and empty cylinder, protection from sunlight, storage in well-ventilated area, storage of different types of cylinders away from each other and providing valve caps to prevent damage to cylinder valves.

78. Concrete handling/pouring and placing

- Prior to pouring the concrete inspect form work & their supports shall be ensured by use of proper checklist.
- Obtain certification from competent personnel/undertaking for safe platform to be given by NOMINATED SUBCONTRACTOR's project in charge.
- The crane operators must be trained for this type of work and must have normal vision. If large distances are covered, two-way telephone communication or walkie-talkies have to be used.
- When concrete pumps with pipelines and placer masts are used, special attention should be paid to the stability of the installation. Agitating Lorries (cement mixers) with built-in concrete pumps must be equipped with interlocked switches which make it impossible to start the two operations simultaneously. The agitators must be guarded so that the operating personnel cannot come into contact with moving parts.
- When agitating Lorries are used in combination with placing plant and lifting equipment, special attention has to be paid to overhead electric lines. Unless the overhead line can be displaced they must be insulated or guarded by protective scaffolds within the work range to exclude any accidental contact. It is important to contact the power supply station.
- Persons working shall wear helmet, gumboots, rubber hand gloves, nose mask, ear plugs and safety goggles to protect him from danger in such handling.
- Keep as much of his body covered by providing suitable clothing for cement handling to minimize contact with cement dust.
- Takes all necessary precautions to keep cement and concrete away from his skin in such handling.
- When concreting is at height, scaffolding with safe access to the working platform protected by guard rails &suitable safety netting arrangements shall be provided to all exposed sides

DRAFT

where there is possibility of fall and workers shall wear safety harness & anchor the life line to firm member/structure.

- Any repair work on the machine shall be carried out after switching off the power supply.
- If the person has to enter inside the drum to clean, then work permit has to be obtained, required precaution to be taken and work has to be carried out under supervision.
- The area of concreting shall be cleaned after concreting work is completed .Also the waste concrete near the pump shall be removed and disposed off.

Pipes and Pumps:

- Scaffolding carrying a pipe for pumped concrete is strong enough to support such pipe at time when such pipe is filled with concrete or water.
- The length of concrete pump pipe line (horizontally) shall be supported with suitable material.
- Vertical pipe line; Sufficient anchorage should be provided for vertical running of pipe.
- Every pipe for carrying pumped concrete is-
    1. Securely anchored at its end points and at each curve.
    2. provided near the top of such pipe with an air release valve and
    3. Securely attached to pump nozzle by a bolted collar or other adequate means.
- After the concreting where passing or removing the chock, final traces of concrete the ball is pass through the pipeline by air pressure. At this stage the open end of the pipeline should be provided blocker (Guard cap) to prevent the concrete, ball hitting somebody and also sufficient warning should be given to other workers working around to evacuate area during operation.
- Pneumatic lines / compressor etc. should be maintained in good working condition.
- Handling of concrete pipeline shifted carefully at elevated level on the slab and lowered carefully without dropping on steel reinforcement &shuttering for avoiding the bouncing of pipeline and dropping below on the worker working on the ground level.
- Good house-keeping should be maintained the area of movement around concrete pump should be free of obstruction.
- The plant effluent of batching plant operation should be collected in a chamber/sump, and pumped out periodically to the effluent treatment.
- Lime pits are fenced or enclosed.
- Lime pits are filled and emptied by such devices which do not require workers to go into the pit.
- Electric equipment should be having proper body earthing securely connected to the ground earthing system.
- Trained operator shall operate concrete pump.
- Mandatory provision of testing of wire ropes, pulleys, brake switches should be checked periodically for any noticeable defects.

79. Late Night working:

Late night working only in the presence of responsible engineers/supervisors for safe execution of work and ensure presence of responsible person during late night work to monitor , control unsafe practices and to protect the safety of the late night workers .Night work permit to be taken for night work.

DRAFT

80. Miscellaneous:

- The site management reserves the right to supplement the site regulations if necessary within the sense of the contract documents.
- In addition to the site regulations the site management will publish site instructions on special occasions and these site instructions shall be observed like site regulations.
- These site regulations shall not relive the NOMINATED SUBCONTRACTOR's from the responsibility.
- Stick supervision at all times at each higher floor.
- Provision of Toilet/medical facilities at higher floors.
- Specific provision of Assembly area at intermediate floor.
- Monitoring of wind speed /provision of vertical sheets for arresting the wind.
- Oxygen level monitoring, screening of people with vertigo, etc.
- Before start of every day work hazard identification and provision of operational controls by site Engineer.

81. Height Pass:

- Height pass for all workers & supervising staff engaged by NOMINATED SUBCONTRACTOR shall be obtained which includes a medical test by qualified doctor by NOMINATED SUBCONTRACTOR sand physical test for all works where these workers/supervisors need to be work above a height of ± 3.5 m, which is mandatory.

DRAFT

ACKNOWEDGEMENT OF SAFETY RULES AND REGULATIONS BY NOMINATED SUBCONTRACTOR / SUB-CONTRACOTRS

AGREEMENT:

Contract Company Name: ----------

Contract /PO Number: ----------

Duration of Work: ----------

Scope of work: ----------

NOMINATED SUBCONTRACTOR ssafety representative: ----------

I (Project Manager Name) ----------Designation---------- of (Company) ----------is being a NOMINATED SUBCONTRACTOR sfor the project at----------, do herby confirm that I have been briefed by the Project Manager (Name) ----------on---------- on safety rules &regulations.

I fully understand the requirements and the requirements communicated to me are clear and I agree to abide by the rules &regulations while deploying our workers at the site. Any contravention of the rules and regulations on our part shall render us,

---------- (Company) fully accountable to all the consequences.

I confirm that I am a Director/Authorized and appointed representative of the Company.

Signed by:

NOMINATED SUBCONTRACTOR Company Stamp
Signature & date

Contact Number: ----------

Witnessed by:

Project Manager
Signature &Date

Contact Number: ----------

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